The National Institute of Engineering, Mysuru

Autonomous Institute, Affliated to VTU

North Campus No.50 (Part), Koorgalli village, Hootagalli Industrial Area, Mysuru-570018

Karnataka

Department of Computer Science and Engineering

Laboratory Record

For

Full Stack Development – BCSL504



Name: Shreeja Hebbar

USN: 4NI24CS201

B.E – 5th SEM

AY: 2026-2027

INDEX

Expt. No	Experiment Title	Page No
5	Illustrate react program to implement error boundaries to find 1. Division by zero error. 2. Array index of out of bound.	

Experiment 5

Illustrate react program to implement error boundaries to find
1. Division by zero error.
2. Array index of out of bound.

Code:

```
<!DOCTYPE html>
<html>
<head>
  <title>React Error Boundary</title>

  <script src="https://unpkg.com/react@18/umd/react.development.js"></script>
  <script src="https://unpkg.com/react-dom@18/umd/react-dom.development.js"></script>
  <script src="https://unpkg.com/@babel/standalone/babel.min.js"></script>

  <style>
    body {
      font-family: Arial;
      text-align: center;
      margin-top: 50px;
    }

    .box {
      border: 2px solid #333;
      padding: 20px;
      margin: 20px auto;
      width: 400px;
    }

    button {
      padding: 10px 20px;
      margin: 10px;
      cursor: pointer;
    }

    .error {
      color: red;
      font-weight: bold;
    }
  </style>
</head>

<body>

  <div id="root"></div>

  <script type="text/babel">

    // Error Boundary Component
    class ErrorBoundary extends React.Component {

      constructor(props) {
```

```
        super(props);

        this.state = {
            hasError: false,
            errorMessage: ""
        };
    }

    static getDerivedStateFromError(error) {
        return {
            hasError: true,
            errorMessage: error.message
        };
    }

    componentDidCatch(error, info) {
        console.log("Error caught:", error);
        console.log("Error details:", info);
    }

    render() {
        if (this.state.hasError) {
            return (
                <div className="box error">
                    <h2>Something went wrong!</h2>
                    <p>{this.state.errorMessage}</p>
                </div>
            );
        }

        return this.props.children;
    }
}

// Division by Zero Component
class DivisionByZero extends React.Component {

    constructor(props) {
        super(props);

        this.state = {
            calculate: false
        };
    }

    handleClick = () => {
        this.setState({ calculate: true });
    };

    render() {

        if (this.state.calculate) {

            // JavaScript normally returns Infinity,
            // so we manually throw an error.
```

```
            let denominator = 0;

            if (denominator === 0) {
                throw new Error("Division by zero is not allowed!");
            }

            return <h3>Result: {10 / denominator}</h3>;
        }

        return (
            <div className="box">
                <h2>Division by Zero</h2>
                <button onClick={this.handleClick}>
                    Divide 10 / 0
                </button>
            </div>
        );
    }
}

// Array Index Out of Bounds Component
class ArrayError extends React.Component {

    constructor(props) {
        super(props);

        this.state = {
            access: false
        };
    }

    handleClick = () => {
        this.setState({ access: true });
    };

    render() {

        const numbers = [10, 20, 30];

        if (this.state.access) {

            let index = 5;

            if (index >= numbers.length) {
                throw new Error(
                    "Array index out of bounds! Index " +
                    index +
                    " does not exist."
                );
            }

            return <h3>{numbers[index]}</h3>;
        }

        return (
```

```
                <div className="box">
                    <h2>Array Index Out of Bounds</h2>
                    <p>Array: [10, 20, 30]</p>

                    <button onClick={this.handleClick}>
                        Access Index 5
                    </button>
                </div>
            );
        }
    }

    // Main App
    class App extends React.Component {

        render() {
            return (
                <div>
                    <h1>React Error Boundary Demo</h1>

                    <ErrorBoundary>
                        <DivisionByZero />
                    </ErrorBoundary>

                    <ErrorBoundary>
                        <ArrayError />
                    </ErrorBoundary>

                </div>
            );
        }
    }

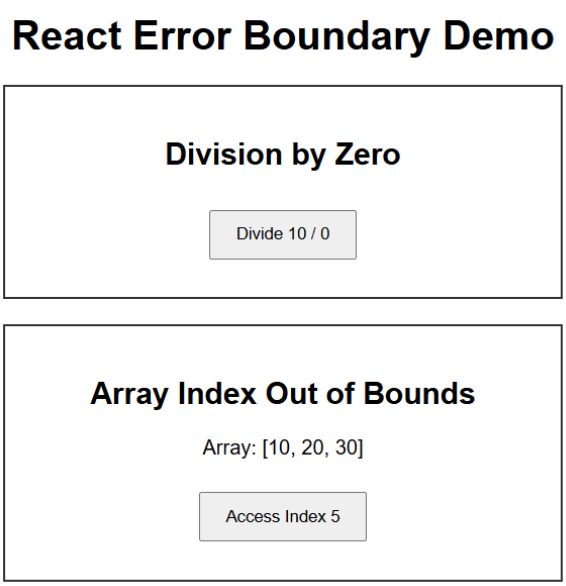
    // Render React Application
    const root = ReactDOM.createRoot(
        document.getElementById("root")
    );

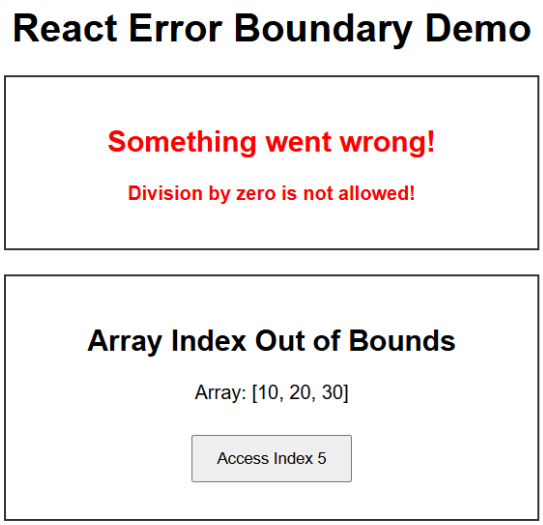
    root.render(<App />);

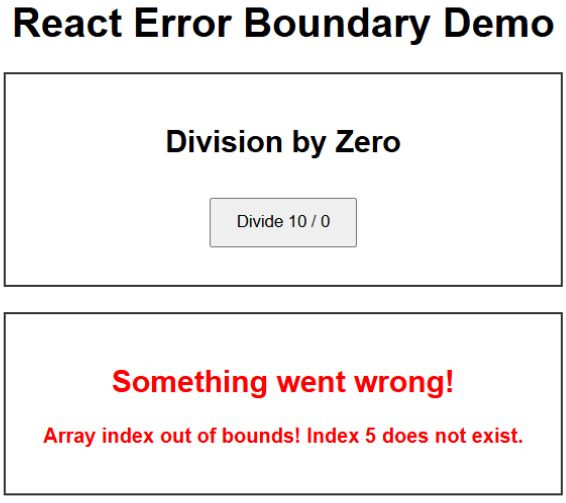
  </script>

</body>
</html>
```

Output:


React Error Boundary Demo
Division by Zero
Divide 10 / 0
Array Index Out of Bounds
Array: [10, 20, 30]
Access Index 5


React Error Boundary Demo
Something went wrong!
Division by zero is not allowed!
Array Index Out of Bounds
Array: [10, 20, 30]
Access Index 5


React Error Boundary Demo
Division by Zero
Divide 10 / 0
Something went wrong!
Array index out of bounds! Index 5 does not exist.